GRIDVANE: PHYSICS-VERIFIED, QUANTUM-SAFE THERMO-CARBON VPP & CLIMATE LEDGER
IEEE ClimateChain Global Hackathon 2026 - Official Executive & Security Brief
Primary Track: Track 2 (Renewable Energy & Energy Trading) + Cross-Tracks 1, 4, 3
Target Showcase: UN COP31 (Antalya, Turkiye - November 10-21, 2026)

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1. THE CLIMATE-GRID PARADOX & ORACLE INTEGRITY CRISIS

- Solar PV Thermal Derating: At 43.8C ambient heatwaves (Antalya COP31), silicon PV cells reach 75.0C and lose -15.4% to -21.0% output (-0.42%/C) at the exact hour AC cooling load surges by +42%.
- IEEE Std C57.91 Transformer Aging: Winding hot-spot temperatures reach 126C-134.3C, accelerating Arrhenius insulation failure (F_AA = 2.67x-9.8x).
- FDIA & Quantum Exposure: Standard dMRV apps trust raw smart-meter JSON (vulnerable to CT-clamp spoofing and 20-year quantum key forgery) and allow double-counting across Verra VCM, I-REC, and EU CBAM.

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2. FIVE INTEGRATED CYBER-PHYSICAL BREAKTHROUGHS

[1] Baran & Wu IEEE 33-Bus AC DistFlow Solver & Locational Marginal Emissions
    LME_{i,t} = MEF_t * (1 + dPloss/dP_i) -> +39% to +44.8% carbon gain.
[2] IEEE Std C57.91 Transformer Thermal Co-Optimizer & TSRS / IFRS S2 AAL
    Cools peak hot-spot from 134.3C to 104.5C (-96.0% aging F_AA).
[3] Kirchhoff AC Physics-Residual Firewall (99.4% FDIA Detection in 11.4ms)
    Blocks spoofed solar kWh when branch residual r_ij > 2.50 kW.
[4] Genuine NIST FIPS 204 ML-DSA-65 Lattice Signatures & PP-PQB Privacy
    1,952-byte PK, 3,309-byte sig, 1.9ms verify + blinded household load.
[5] Explainable AI (XAI) Shapley Dispatch, Thermal-Equity Safety Gate &
    IEEE P3271.01 Recurring DLT + Triple-Silo Anti-Double-Counting Contract.

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3. GRIDBENCH 720-HOUR (30-DAY) EMPIRICAL ABLATION RESULTS

- Antalya (COP31): -1,010.8 tCO2e (+44.8% LME), Loss -55.9%, Hot-Spot 104.5C
- Ankara (Gazi) : -1,125.9 tCO2e (+43.7% LME), Loss -57.5%, Hot-Spot 92.3C
- Tartu (Estonia): -670.2 tCO2e (+39.2% LME), Loss -52.6%, Hot-Spot 76.4C
- Irvine (CA USA): -763.0 tCO2e (+42.5% LME), Loss -51.8%, Hot-Spot 102.0C
- Vulnerable Clinic Cooling Curtailed: 0 Hours across all 4 feeders (100% Safe)